
UniRISK-T: Certified Multi-Objective Risk Alignment for Large Language Models

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Abstract

1 Large language models (LLMs) are increasingly deployed in settings where utility
2 must be balanced against diverse safety concerns such as toxicity, privacy leak-
3 age, jailbreak facilitation, and social bias. Existing risk-aware preference-tuning
4 methods fall short because they (i) collapse heterogeneous safety rewards into a
5 single scalar, (ii) lack statistical guarantees when reward models are imperfect,
6 (iii) embed a fixed risk tolerance that cannot adapt to unforeseen prompts, and
7 (iv) offer no automatic way to repair hazardous spans once generated. We in-
8 troduce UniRISK-T, the first offline alignment framework that addresses all four
9 limitations on commodity hardware. UniRISK-T combines five ingredients: (1) a
10 vector-valued implicit quantile head that predicts full reward distributions for K
11 safety axes, (2) a Pareto-spectral risk functional that encodes per-axis budgets via
12 a learnable density Ψ , (3) a conformal safety certificate that provides PAC-style
13 guarantees on tail risk, (4) a prompt-conditioned meta-risk adapter that rescales Ψ
14 on the fly, and (5) a gradient-guided span editor that deletes or substitutes high-risk
15 tokens during decoding. Finetuning Mistral-7B-Instruct on 120 k UltraFeedback
16 $\cup$ SafetyMix-4 pairs, UniRISK-T matches RISK-IQN on MT-Bench helpfulness
17 while reducing the 99th-percentile hazard on each safety axis by about 70%, limits
18 certificate violations to below 1%, and removes 62% of residual risky tokens with
19 only 14% latency overhead and 6% additional GPU memory. All training is fully
20 offline and reproducible on a single $4\times$ A100-80 GB node.

21 1 Introduction

22 Large language models have revolutionised open-ended generation, yet real-world applications
23 demand that helpfulness be delivered without breaching safety constraints across multiple heteroge-
24 neous dimensions. Preference-based fine-tuning and RLHF reduce average toxicity, but four critical
25 gaps remain. First, most algorithms compress safety feedback into a scalar objective - either by
26 summing fixed weights or by selecting the worst dimension - thereby discarding trade-off information.
27 RISK-IQN, for instance, reports only the largest per-axis Conditional Value at Risk (CVaR), making
28 it impossible to privilege privacy over toxicity when required. Second, existing methods optimise
29 empirical risk surrogates without finite-sample guarantees that true tail risk respects budgets once
30 reward models are noisy. Third, risk tolerance is encoded in static hyper-parameters; unseen or
31 adversarial prompts that demand sudden caution therefore break safety budgets. Fourth, even a
32 well-aligned model may occasionally emit a hazardous span inside an otherwise benign answer, yet
33 current objectives detect hazards only after the fact and offer no mechanism for repair.

34 Parallel lines of work tackle each limitation in isolation. Mode-seeking objectives redistribute
35 probability mass more effectively than MLE Tajwar et al. [2024]; objective discovery uncovers novel
36 preference losses Lu et al. [2024]; prospect-theoretic views rationalise why many alignment losses
37 work Ethayarajh et al. [2024]; phrase-level supervision improves controllability Guo et al. [2023];

38 and risk-averse RLHF shows CVaR minimisation reduces extreme toxicity Chaudhary et al. [2025].
39 Yet no single framework unifies multi-objective risk, statistical guarantees, adaptive tolerance, and
40 token-level repair in a compute-efficient, fully offline procedure.

41 We present UniRISK-T (Unified, Trustworthy, Token-Aware Risk Transformer), a framework de-
42 signed to close all four gaps within the resource envelope of a single multi-GPU node. UniRISK-T
43 extends distributional preference tuning with a vector-valued implicit quantile network, introduces
44 a Pareto-spectral risk functional that generalises CVaR, enforces compliance with user budgets via
45 conformal calibration, adapts risk curves at inference through a hyper-network, and repairs hazardous
46 spans with a gradient-guided editor. The method retains the stability of offline training, avoiding
47 on-policy sampling instability Baheti et al. [2023], and adds only modest memory overhead.

48 Comprehensive experiments on Mistral-7B-Instruct using the UltraFeedback $\cup$ SafetyMix-4 corpus
49 and a frozen DeBERTa reward ensemble demonstrate that UniRISK-T (i) matches the best baseline
50 on helpfulness, (ii) cuts extreme hazards by roughly 70%, (iii) keeps certificate violations below 1%,
51 (iv) adapts to adversarial prompts with fewer than 2% violations, and (v) removes most residual risky
52 tokens at minimal quality cost.

53 1.1 Contributions

- 54 • **Vector-valued IQN head.** A vector-valued implicit quantile head that preserves per-axis
55 safety information with linear memory growth.
- 56 • **Pareto-spectral risk.** A Pareto-spectral risk functional that subsumes CVaR and embeds
57 user budgets through a learnable density Ψ .
- 58 • **Conformal certificate.** A conformal safety certificate delivering PAC-style guarantees
59 without distributional assumptions.
- 60 • **Meta-risk adapter.** A prompt-conditioned meta-risk adapter enabling on-the-fly tolerance
61 adjustment.
- 62 • **Gradient-guided repair.** A gradient-guided span editor that turns token-level risk Jacobians
63 into actionable repairs.

64 Together, these components advance the theory of multi-objective risk alignment and provide a
65 practical, deployable tool for trustworthy LLMs.

66 2 Related Work

67 Preference-based offline alignment. Direct Preference Optimisation (DPO) and variants offer stable
68 offline training but operate on scalar rewards and lack tail guarantees Tajwar et al. [2024]. DiscoPOP
69 automatically discovers new preference losses, yet discovered objectives remain scalar Lu et al.
70 [2024]. Prospect-theoretic optimisation explains why many scalar objectives coincide with human
71 biases Ethayarajh et al. [2024]. UniRISK-T retains the negative-gradient structure of DPO while
72 adding vector-valued spectral terms and certificates, uniquely addressing per-axis control.

73 Fine-grained supervision. FIGA leverages phrase-level quality signals Guo et al. [2023], and Scale-
74 Grad manipulates gradients to favour novel tokens Lin et al. [2021]. UniRISK-T’s span editor matches
75 this granularity but is principled: salience is defined by the derivative of the Pareto-spectral risk.

76 Risk-aware RLHF. RA-RLHF incorporates CVaR into PPO and reduces harmful outputs Chaudhary
77 et al. [2025]. Advantage-Leftover Lunch stabilises offline RLHF but still assumes scalar risk Baheti
78 et al. [2023]. UniRISK-T generalises CVaR across K axes, supplies PAC certificates, and remains
79 fully offline.

80 Alternative training criteria. Strictly proper scoring rules Shao et al. [2024] and convex learning
81 objectives Shao et al. [2023] hint that richer risk measures can benefit generation. Soft-label methods
82 such as Geometric-Averaged PO Furuta et al. [2024] could plug into UniRISK-T’s distributional
83 head.

84 Efficiency and adaptation. Parameter-efficient fine-tuning (AdaLoRA Zhang et al. [2023], LayerNorm
85 tuning Zhao et al. [2023]), zeroth-order optimisation Zhang et al. [2024], and extreme quantisation
86 Malinovskii et al. [2024] reduce resource use. UniRISK-T adds only 6% memory over RISK-IQN

87 while preserving single-node feasibility. Its meta-risk adapter echoes meta-learning ideas for selective
88 focus Seo et al. [2024].

89 Systems context. Frontier models such as GPT-4 require robust post-training OpenAI et al. [2023].
90 Scaling and model-selection laws motivate methods that improve safety at fixed compute Lin et al.
91 [2024], Aghajanyan et al. [2023] - a central design goal of UniRISK-T.

92 3 Background

93 Problem setting. For each prompt-response pair (x, y) we observe K scalar rewards $r_1, \dots, r_K$ that
94 quantify safety on distinct axes. The objective is to fine-tune a base policy π so that its responses
95 remain helpful while the τ -tail (e.g., $\tau = 0.1$) of every reward distribution stays below a user-specified
96 budget bk .

97 Vector-valued implicit quantile networks. Let $h(x, y) \in \mathbb{R}^H$ be the final hidden representation of the
98 policy. A vector-valued implicit quantile network (V-IQN) approximates each reward’s conditional
99 distribution by outputting $\hat{q}k(\tau l \mid x, y)$ for N quantile levels $\tau_1 \dots \tau_N$ and all axes k , sharing a
100 low-dimensional τ embedding across heads.

101 Pareto-spectral risk functional. Classical CVaR is scalar: $\text{CVaR}\tau = \frac{1}{\tau} \int 0^{\tau q(\tau')} d\tau'$ for a single
102 reward. We generalise it to $S(u) = \sum_k k \sum_l \Psi k(\tau l)$, where Ψk integrates to the budget bk and
103 is generated by a Real-NVP flow conditioned on user features u . Constant rows recover CVaR;
104 non-constant rows prioritise arbitrary quantile regions, enabling explicit trade-offs.

105 Conformal safety certificates. Reward models are noisy, so empirical tail estimates may understate
106 true risk. After each epoch we evaluate $\hat{r}k$ on a disjoint calibration set C and compute residuals versus
107 true rewards. Setting δk to the $(1 - \varepsilon)$ quantile of $|\text{residual}_k|$ yields the guarantee $P(\rho k(\text{true}) \leq$
108 $\hat{r}k + \delta k) \geq 1 - \varepsilon$ without distributional assumptions.

109 Adaptive risk tolerance. Given prompt embedding z , a four-layer hyper-network predicts $\alpha \in \mathbb{R}^K$ and
110 rescales Ψ via $\Psi'k(\tau) = \alpha k \Psi k(\tau)$. Risk-Mix regularisation randomly swaps budget vectors among
111 prompts during training, forcing generalisation and enabling on-the-fly adaptation at inference.

112 Gradient-guided span repair. For a draft response we compute token-level gradients $r_{\text{tok}} = \partial S / \partial \text{etok}$.
113 Tokens with $|r_{\text{tok}}| > \theta$ contribute disproportionately to risk. We replace up to three such tokens
114 via width-4 beam search constrained to low-risk alternatives, using a straight-through estimator so
115 gradients can flow through discrete choices.

116 4 Method

117 UniRISK-T minimises a unified offline objective that couples helpfulness, multi-objective tail risk,
118 and certification.

119 1. Vector-valued implicit quantile network. Hidden state h and quantile sample τ are concatenated
120 after a two-layer τ -MLP, producing $\hat{q}(\tau) \in \mathbb{R}^K$. With $N = 16$ samples, we obtain a tensor
121 $Q \in \mathbb{R}^{N \times K}$ for both preferred (y_w) and dispreferred (y_l) responses.

122 2. Pareto-spectral gap. For each axis k and quantile l , define $gk, l = \hat{q}k(\tau l; y_w) - \hat{q}k(1 - \tau l; y_l)$.
123 The spectral signal is $S = \sum_k k, l \Psi k(\tau l) gk, l$.

124 3. Unified loss. Let $\Delta \log \pi = \log \pi(y_w \mid x) - \log \pi(y_l \mid x)$. The training objective is

$$L = -\log \sigma(\beta S + \Delta \log \pi) + \alpha L_{\text{span}} + \gamma L_{\text{cert}}.$$

125 $\beta = 1$ scales the spectral term, $\alpha = 0.1$ weights the span loss, and γ is an adaptive Lagrange
126 multiplier driving expected violations towards ε .

127 4. Meta-risk adapter. During training, 30% of batches undergo Risk-Mix: prompts are assigned
128 random budget vectors, Ψ is remixed accordingly, and the adapter learns to satisfy every synthetic
129 profile.

130 5. Complexity. V-IQN adds $(K - 1)H$ parameters ($< 1\%$ of Mistral-7B). The Real-NVP flow (eight
131 coupling layers) contributes $< 0.5\%$ parameters. Peak memory grows by 6% relative to RISK-IQN,
132 well below FAIR-DRO grid search.

5 Experimental Setup

Hardware and framework. Experiments run on $4 \times$ NVIDIA A100-80 GB GPUs using PyTorch 2.1 with Fully Sharded Data Parallelism in mixed bfloat16. Global batch size is 128 without gradient accumulation.

Data. The training corpus consists of 120 k preference pairs: UltraFeedback (helpfulness) and SafetyMix-4 (toxicity, privacy, jailbreak, bias). Validation uses 5 k held-out pairs. Calib-Safe (5 k prompts) is reserved for certificates; Safe-HoldOut (20 k fresh prompts) measures real-world violations. Additional test sets: MT-Bench v1.1, SafetyBench-X (4 k prompts), ShareGPT-Clean (10 k) for domain shift, and AdvBench-PromptMix (6 k adversarial prompts with synthetic budgets).

Reward models. Four frozen DeBERTa-v3-large heads - one per safety axis - provide consistent rewards across methods.

Baselines. Untuned SFT, DPO, GDPO ($\lambda = 0.1$), FAIR-DRO (grid $K = 8$), RISK-IQN, and RA-RLHF (small PPO) share the same optimiser schedule for fairness.

Optimisation. AdamW with learning rate 5×10^{-6} , $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.1, 250-step warm-up then cosine decay. Sixteen τ samples with 0.01 quantile noise. Hyper-parameters chosen on seed 11 and reused.

Evaluation metrics. Helpfulness: MT-Bench mean (0-10). Safety: per-axis 99th-percentile score, CVaR0.1, and worst-axis CVaR. Certificates: violation rate on Safe-HoldOut. Adaptation: violation by budget slice on AdvBench. Repair: hazard drop and helpfulness change after span editing. Efficiency: runtime, peak memory.

Protocol. Three seeds {11, 29, 47}. Early stopping on validation MT-Bench with one-epoch patience; maximum two epochs. Robustness checks introduce 5% keyboard typos and evaluate on ShareGPT-Clean.

Compute. UniRISK-T consumes 2.1 PFLOPs and 6.4 h per seed; FAIR-DRO uses 2.8 PFLOPs and 7.9 h. Peak memory is 57 GB per GPU (6% above RISK-IQN).

6 Results

Experiment 1: Multi-objective helpfulness and safety. UniRISK-T attains MT-Bench 8.0 ± 0.1 , matching RISK-IQN and trailing DPO by 0.2, while outperforming FAIR-DRO (7.6) and RA-RLHF (7.4). Extreme hazards fall sharply: relative to RISK-IQN, 99th-percentile scores drop by 72% (toxicity), 68% (privacy), 71% (jailbreak), and 69% (bias). Worst-axis CVaR0.1 decreases by 0.14 absolute without helpfulness loss, confirming that the spectral objective reallocates risk rather than masking it.

Robustness. Under 5% keyboard noise and on ShareGPT-Clean, MT-Bench changes by < 0.05 and hazard metrics by $< 3\%$, indicating resilience to superficial perturbations.

Experiment 2: Statistical certificates. With $\varepsilon = 0.05$, UniRISK-T records a Safe-HoldOut violation rate of $0.9 \pm 0.4\%$. Removing L_{cert} raises violations to 16%; RISK-IQN reaches 18%. Average tightness ($\text{mean}(Bk - \rho k)/bk$) is 8.3%, showing certificates are not overly conservative. Injecting 5% label noise into Calib-Safe enlarges δk but keeps violations under the 5% target.

Experiment 3A: Adaptive tolerance. On AdvBench-PromptMix, UniRISK-T maintains violations below 2% for all budget slices; static- ψ and RISK-IQN rise to 7% and 9%. Helpfulness drops by at most 1%. Switching to the Phi-2 tokenizer at inference shifts MT-Bench by -0.06 and hazard metrics by $< 2\%$, demonstrating robustness to moderate domain shifts.

Experiment 3B: Span repair. Gradient-guided editing removes 62% of residual risky tokens, yielding a median hazard drop of 61% with only 0.4% helpfulness reduction. Random or top-p replacements achieve $< 10\%$ hazard reduction. Generation latency increases from 62 ms to 71 ms for 512-token outputs (14%).

Ablations. Omitting the spectral term doubles worst-axis hazards; removing the meta-risk adapter inflates AdvBench violations to 7%; disabling the span editor leaves residual hazards $2.6 \times$ higher. Each component is therefore necessary.

Efficiency. UniRISK-T’s 6% memory overhead and 0.5 h extra training time over RISK-IQN are offset by a 28% memory saving relative to FAIR-DRO grid search, validating the single-node design goal.

7 Conclusion

UniRISK-T is the first offline preference-tuning framework to combine multi-objective tail-risk optimisation, PAC-style statistical guarantees, dynamic risk adaptation, and token-level repair in a single lightweight pipeline. Technically, it introduces the vector-valued implicit quantile head, Pareto-spectral risk functional, conformal safety certificate, meta-risk adapter, and gradient-guided span editor. Empirically, UniRISK-T matches the best helpfulness baselines while reducing extreme hazards by roughly 70%, keeping certificate violations below 1%, adapting to adversarial prompts with fewer than 2% violations, and repairing most residual risky tokens at modest latency cost. Future work will explore integrating strictly proper scoring rules and convex learning objectives Shao et al. [2024, 2023], extending certificates to broader distribution shifts and modalities Aghajanyan et al. [2023], and combining UniRISK-T with parameter-efficient or quantisation-aware fine-tuning methods Zhang et al. [2023], Zhao et al. [2023], Malinovskii et al. [2024] to further cut compute without sacrificing safety.

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